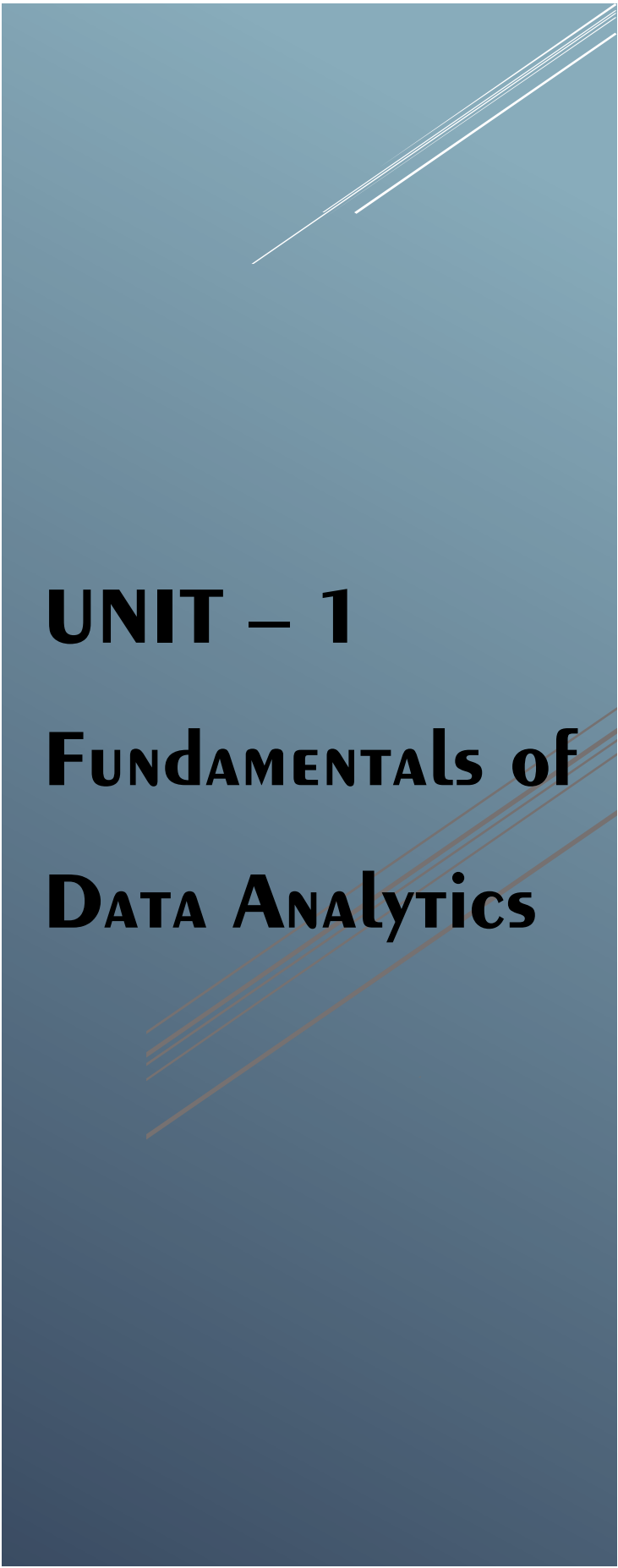




UNIT – 1

FUNDAMENTALS of

DATA ANALYTICS

602
Data Analytics
using
Python

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1.1 Exploratory Data Analysis:

- Exploratory Data Analysis, EDA, is the first step in understanding a dataset.
- Before we build any machine learning model, we must explore the data to:
 - Understand what's in it
 - Find patterns or trends
 - Spot mistakes, missing values, or unusual data (outliers)
- Exploratory Data Analysis (EDA) can be divided into three main types based on the number of variables being analysed:
 - Univariate Analysis
 - Bivariate Analysis
 - Multivariate Analysis

1.1.1 Univariate Analysis:

- Univariate analysis is the simplest form of data analysis where **only one variable** is analyzed at a time.
- The primary goal is to **describe the distribution, central tendency, and variability** of that variable, without considering relationships with other variables.
 - **"Uni-"** means one → only one variable is studied.
 - Focuses on **summary statistics** and **visualizations** to understand patterns in the data.
- **Example:**
 - Analyzing the ages of students in a classroom.
 - Counting how many times each product was sold in a store.
- **Objectives of Univariate Analysis**
 - **Understand the data distribution** – see how values are spread.
 - **Summarize the variable** – identify central tendency (mean, median, mode).

- **Detect outliers or anomalies** – unusual values that could affect analysis.
- **Assess variability** – how much the data varies.
- **Make decisions about data pre-processing** – e.g., whether normalization or transformation is needed.

Types of Variables in Univariate Analysis:

- Quantitative (Numerical) Variables
 - Variables that can be measured or counted.
 - **Continuous:** can take any value in a range (e.g., height, weight, temperature).
 - **Discrete:** countable values (e.g., number of students, number of cars).
- Qualitative (Categorical) Variables
 - Variables that describe categories or groups.
 - **Nominal:** categories without order (e.g., gender, color, city).
 - **Ordinal:** categories with an order (e.g., ratings: low, medium, high).

Example:

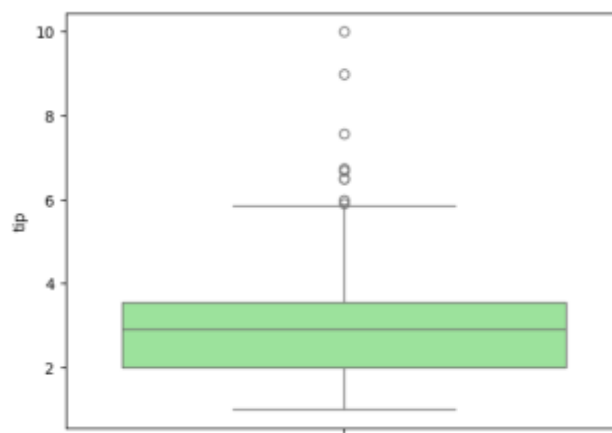
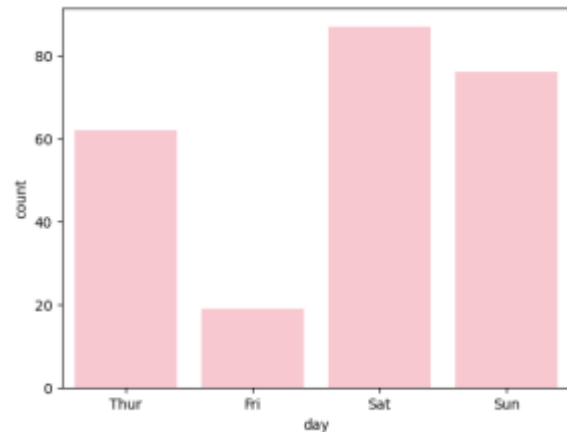
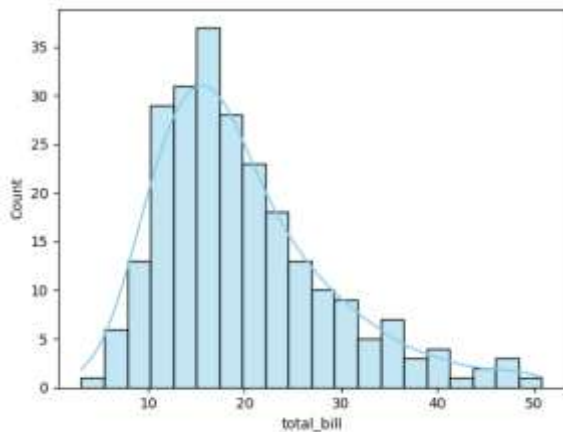
```
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = sns.load_dataset('tips')

# Histogram of the 'total_bill' column
sns.histplot(df['total_bill'], bins=20, kde=True, color='skyblue')
plt.show()

# Bar graph of the 'day' column
sns.countplot(x=df['day'], color='pink')
plt.show()

# Boxplot of the 'tip' column
sns.boxplot(y=df['tip'], color='lightgreen')
plt.show()
```



1.1.2 Bivariate Analysis:

- Bivariate analysis is the statistical analysis of **two variables simultaneously** to determine **relationships, associations, or correlations** between them.
 - “**Bi-**” means two → two variables are studied together.
 - Unlike univariate analysis, the focus is **not just on individual variables** but on how one variable **affects or relates** to the other.
- Example:
 - Examining whether **study hours** affect **students’ marks**.
 - Checking if **gender** is related to **preference for a product**.
- Objectives of Bivariate Analysis
 - **Identify relationships** between two variables (positive, negative, or none).

- **Measure strength of association** (e.g., correlation coefficient).
- **Detect trends or patterns** (e.g., one variable increases as the other increases).
- **Predict values** (used in regression analysis).
- **Compare groups** (e.g., average scores between males and females).

Types of Variables in Bivariate Analysis

- The method depends on the **type of the two variables**:

Variable 1	Variable 2	Common Bivariate Techniques
Numerical	Numerical	Correlation, Scatter Plot, Regression
Numerical	Categorical	t-test, ANOVA, Boxplot, Mean Comparison
Categorical	Categorical	Chi-square test, Contingency Table, Bar Plot

Example:

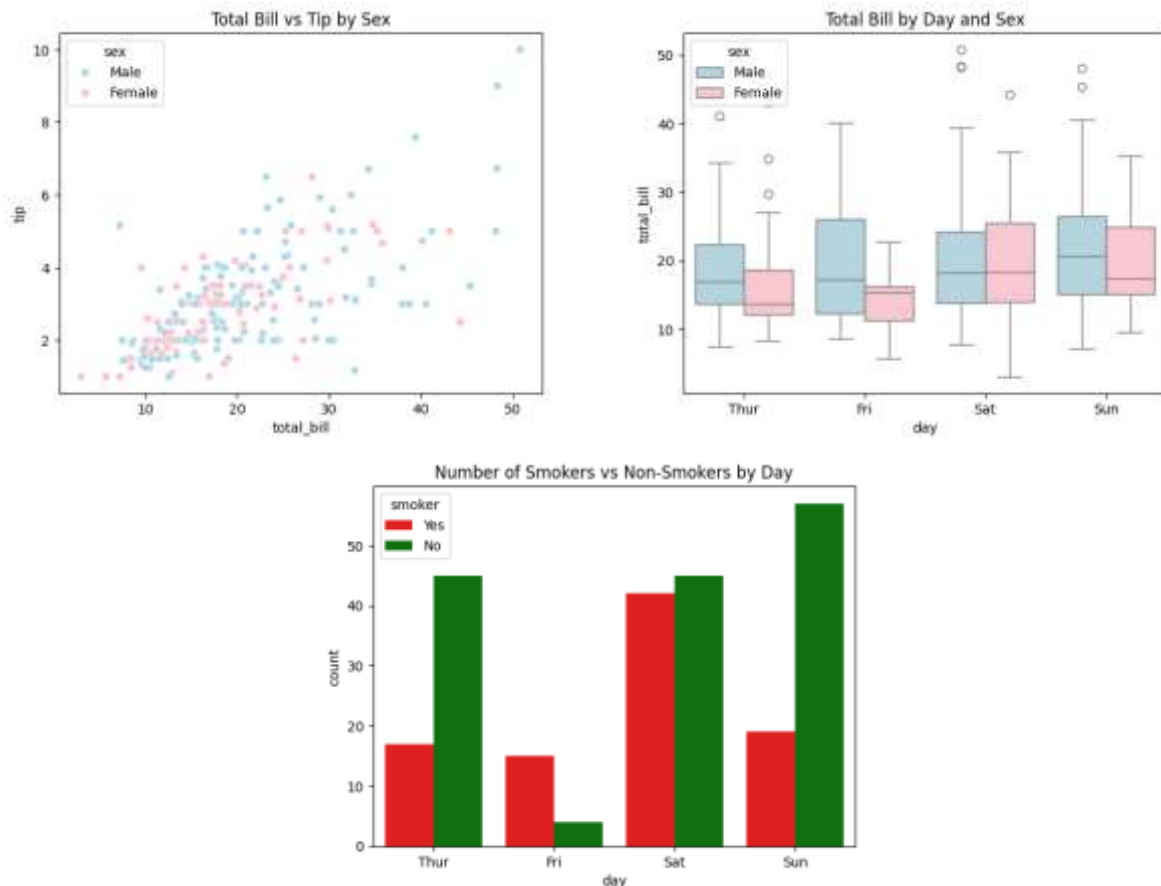
```
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
df = sns.load_dataset('tips')

# Numeric vs Numeric: Scatterplot (total_bill vs tip)
sns.scatterplot(data=df, x='total_bill', y='tip', hue='sex',
palette={'Male':'lightblue','Female':'pink'})
plt.title('Total Bill vs Tip by Sex')
plt.show()

# Categorical vs Numeric: Boxplot (day vs total_bill)
sns.boxplot(data=df, x='day', y='total_bill', hue='sex',
palette={'Male':'lightblue','Female':'pink'})
plt.title('Total Bill by Day and Sex')
plt.show()

# Categorical vs Categorical: Countplot (day vs smoker)
sns.countplot(data=df, x='day', hue='smoker', palette={'Yes':'red','No':'green'})
plt.title('Number of Smokers vs Non-Smokers by Day')
plt.show()
```



1.1.3 Multivariate Analysis:

- Multivariate analysis is the statistical analysis of **three or more variables simultaneously** to understand relationships, patterns, and dependencies among them.
 - “**Multi-**” means many → more than two variables are studied together.
 - It allows **simultaneous examination** of multiple variables, making it more complex than univariate or bivariate analysis.
- Example:
 - Examining how study hours, sleep hours, and attendance together affect exam scores.
 - Predicting house prices using variables like location, area, number of bedrooms, and age of the building.

- **Objectives of Multivariate Analysis**

- **Identify relationships** among multiple variables simultaneously.
- **Understand patterns** and interactions between variables.
- **Predict outcomes** using multiple predictors.
- **Reduce dimensionality** (simplify complex datasets).
- **Control for confounding variables** in analysis.

Example:

```
import seaborn as sns
import matplotlib.pyplot as plt

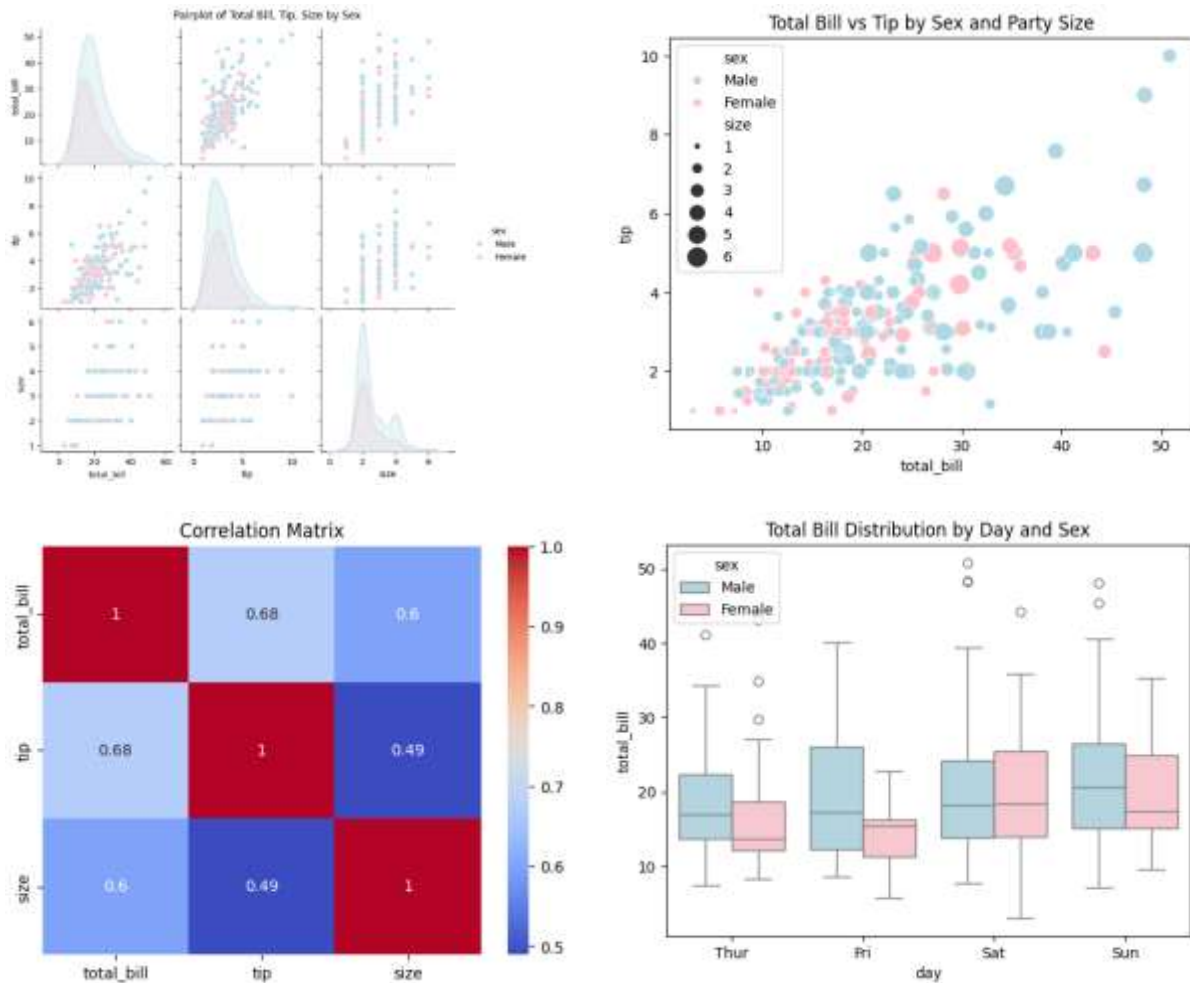
# Load dataset
df = sns.load_dataset('tips')

# Pairplot with 3 variables: total_bill, tip, sex
sns.pairplot(df, vars=['total_bill', 'tip', 'size'], hue='sex',
palette={'Male': 'lightblue', 'Female': 'pink'})
plt.suptitle('Pairplot of Total Bill, Tip, Size by Sex', y=1.02)
plt.show()

# Scatterplot with 3 variables: total_bill vs tip, colored by sex, sized by party size
sns.scatterplot(data=df, x='total_bill', y='tip', hue='sex', size='size',
palette={'Male': 'lightblue', 'Female': 'pink'}, sizes=(20, 200))
plt.title('Total Bill vs Tip by Sex and Party Size')
plt.show()

# Heatmap of correlations among numeric variables
corr = df[['total_bill', 'tip', 'size']].corr()
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Matrix')
plt.show()

# Boxplot: total_bill by day and sex (categorical vs numeric vs categorical)
sns.boxplot(data=df, x='day', y='total_bill', hue='sex',
palette={'Male': 'lightblue', 'Female': 'pink'})
plt.title('Total Bill Distribution by Day and Sex')
plt.show()
```



1.1.4 Handling Missing Data and Outliers:

- In real-world datasets, data is often **incomplete or noisy**. Two common data quality problems are:
 - **Missing data** – some values are not available
 - **Outliers** – extreme values that differ significantly from most data
- Handling them correctly is essential for building **accurate and reliable machine learning models**.

A. Handling Missing Data

What is Missing Data?

- Missing data occurs when one or more values in a dataset are **absent** (e.g., NaN, None, empty).

- **Reasons for missing data:**
 - Data not collected
 - Sensor or system failure
 - Human error during data entry
 - Privacy or non-response issues

Types of Missing Data

- **MCAR (Missing Completely at Random)**
 - Missing values have no relation to other data.
- **MAR (Missing at Random)**
 - Missing values depend on other observed variables.
- **MNAR (Missing Not at Random)**
 - Missing values depend on the missing value itself.

Common Techniques to Handle Missing Data:

1. Removing Missing Data

- Delete rows or columns with missing values.
- Suitable only when missing data is very small.
- **Disadvantages:**
 - Loss of information
 - May introduce bias

2. Imputation (Recommended Approach)

- Replace missing values with estimated values.
- **Types of imputation:**
 - **Mean imputation** – average value (numeric data)
 - **Median imputation** – robust to outliers
 - **Mode imputation** – most frequent value (categorical data)
 - **Constant value imputation** – e.g., 0 or "Unknown"

Example:

```
# Step 1: Import required libraries
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

# Step 2: Load the planets dataset
df = sns.load_dataset('planets')

# Display first 5 rows of the dataset
print("First 5 rows of the dataset:")
print(df.head())

# Step 3: Check dataset information
print("\nDataset Information:")
df.info()

# Step 4: Check for missing values
print("\nMissing values in each column:")
print(df.isnull().sum())

# Step 5: Visualize missing data (Heatmap)
sns.heatmap(df.isnull(), cbar=False)
plt.title("Missing Data Visualization - Planets Dataset")
plt.show()

# Step 6: Select numerical columns for outlier analysis
# (Only numeric columns are selected)
numeric_cols = ['orbital_period', 'mass', 'distance', 'year']

# Step 7: Visualize outliers using boxplot
sns.boxplot(data=df[numeric_cols])
plt.title("Outlier Detection using Boxplot - Planets Dataset")
plt.show()

# Step 8: Handle missing values (Median Imputation)
# Median is preferred because the dataset contains outliers
```

```
df[numeric_cols] = df[numeric_cols].fillna(df[numeric_cols].median())

# Step 9: Handle outliers using IQR method
Q1 = df[numeric_cols].quantile(0.25)
Q3 = df[numeric_cols].quantile(0.75)
IQR = Q3 - Q1

# Remove rows containing outliers
df_cleaned = df[
    ~((df[numeric_cols] < (Q1 - 1.5 * IQR)) |
      (df[numeric_cols] > (Q3 + 1.5 * IQR))).any(axis=1)
]

# Step 10: Compare dataset shape before and after cleaning
print("\nOriginal dataset shape:", df.shape)
print("Cleaned dataset shape:", df_cleaned.shape)

# Step 11: Re-visualize outliers after cleaning
sns.boxplot(data=df_cleaned[numeric_cols])
plt.title("Outliers After IQR Cleaning - Planets Dataset")
plt.show()
```

1.2 Understanding the Data

Understanding the data is the **first and crucial step in data analysis**.

It involves identifying **what type of data you have**, its properties, and how it can be used in analysis.

Data can be broadly classified into **quantitative** and **qualitative** types.

1.2.1 Quantitative Data (Numerical Data)

- Quantitative data is data that can be measured and expressed numerically.
- These data are suitable for mathematical operations like addition, subtraction, mean, standard deviation, etc.

a) Discrete Data

- Data that can take **only specific, separate values**.

- Usually **countable numbers** (integers).
- **Examples:**
 - Number of students in a class (10, 20, 30 ...)
 - Number of cars in a parking lot (0, 1, 2 ...)
- **Characteristics:**
 - No fractions/decimals between values.
 - Often used in **counting** scenarios.

b) Continuous Data

- Data that can take **any value within a range**.
- Usually **measurable quantities**.
- **Examples:**
 - Height of students (160.5 cm, 161.2 cm...)
 - Temperature (36.5°C, 36.7°C ...)
 - Weight (55.3 kg, 60.8 kg ...)
- **Characteristics:**
 - Can have decimals or fractions.
 - Often used in **measurements**.
- **Visualization:**
 - Histograms, line charts, scatter plots.

Type	Description	Examples	Values
Discrete	Countable, separate values	Number of students	0, 1, 2, ...
Continuous	Measurable, any value range	Height, Weight, Temp	5.4, 6.7...

1.2.2 Qualitative Data (Categorical / Non-numerical Data)

- Qualitative data describes **attributes, characteristics, or categories**. These data are **non-numerical** and used for labeling or grouping.

a) Nominal Data (Normal / Non-ordered)

- Categories **without any order or ranking**.

- Used to **label or name** items.
- **Examples:**
 - Gender (Male, Female)
 - Blood Group (A, B, AB, O)
 - Colors (Red, Blue, Green)
- **Characteristics:**
 - Cannot perform arithmetic operations.
 - Analysis focuses on **frequency or proportion**.
- **Visualization:**
 - Bar chart, Pie chart

b) Ordinal Data (Ordered Categories)

- Categories **with a meaningful order**, but differences between them are **not necessarily uniform**.
- **Examples:**
 - Customer satisfaction (Low, Medium, High)
 - Education Level (High School, Graduate, Postgraduate)
 - Ratings (1 star, 2 stars, 3 stars)
- **Characteristics:**
 - Order matters, but exact difference is unknown.
 - Analysis often uses **median, mode, frequency**, not mean.
- **Visualization:**
 - Bar chart, stacked bar chart

Type	Description	Examples	Order/Rank
Nominal	Categories, no order	Gender, Colors	No
Ordinal	Categories with order	Ratings, Education Level	Yes

1.3 Spread of Data

- The **spread of data** refers to how the values of a variable are **distributed or scattered**.
- It helps us understand **variability, patterns, and shape of the data distribution**.
- Spread is crucial for statistical analysis because it affects measures like **mean, median, standard deviation**, and helps determine **which statistical methods are appropriate**.
- Key components in understanding spread:
 - **Distribution shape** → Normal, skewed, etc.
 - **Skewness** → Symmetry of data
 - **Kurtosis** → Peakedness or flatness of data

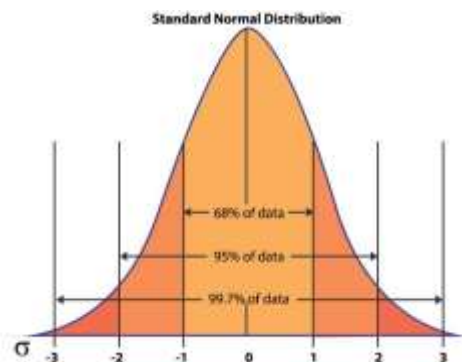
1.3.1 Normal Distribution

- A normal distribution is a **symmetric, bell-shaped distribution** where most values cluster around the **mean**, and probabilities taper off equally on both sides.

Characteristics:

- Symmetrical about the mean.
- Mean = Median = Mode.
- About 68% of values lie within **1 SD** of mean, 95% within **2 SD**, 99.7% within **3 SD** (Empirical Rule). (SD = standard deviation)
- No skewness (Skewness = 0).

Graphical Representation:



Example:

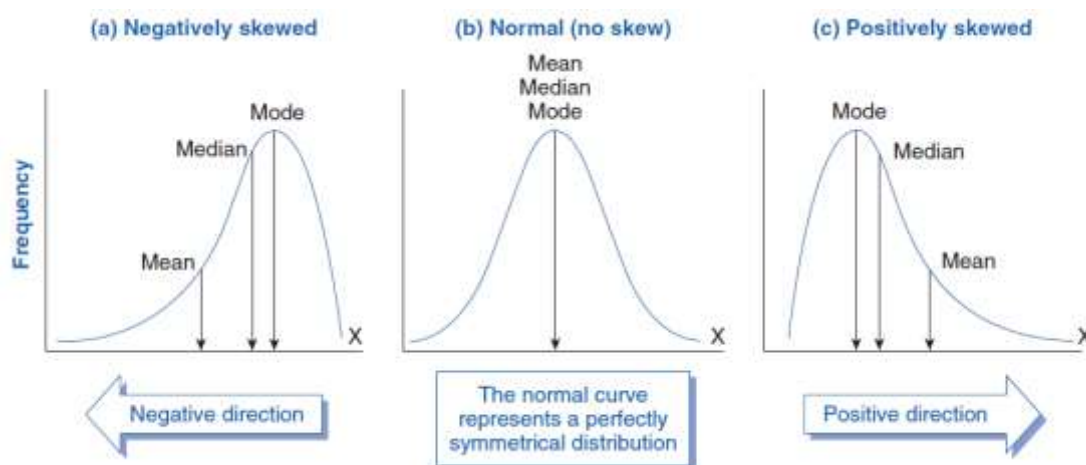
- Heights of adults, IQ scores, measurement errors in manufacturing.

1.3.2 Skewed Distribution

- A skewed distribution is **asymmetric**, where the data values are **concentrated on one side** and tail extends to the other.

Types:

- **Positive Skew (Right Skew)**
 - Tail extends to the **right** (higher values).
 - $\text{Mean} > \text{Median} > \text{Mode}$.
 - Example: Income distribution (few people earn very high amounts).
- **Negative Skew (Left Skew)**
 - Tail extends to the **left** (lower values).
 - $\text{Mean} < \text{Median} < \text{Mode}$.
 - Example: Age at retirement (most retire around 60, few retire early).



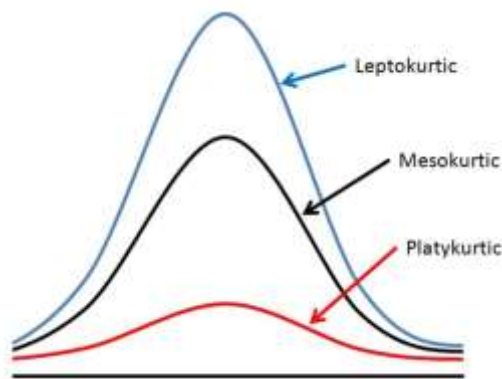
1.3.3 Skewness and Kurtosis

Skewness

- Measures the **asymmetry** of the data distribution.
- **Interpretation:**
 - Skewness = 0 → Symmetrical (normal).
 - Skewness > 0 → Positive skew.
 - Skewness < 0 → Negative skew.

Kurtosis

- Measures the **peakedness or flatness** of the distribution compared to normal distribution.
- **Types:**
 - **Mesokurtic** → Normal peak.
 - **Leptokurtic** → Sharper peak, heavy tails.
 - **Platykurtic** → Flatter peak, light tails.

**Importance:**

- Skewness and kurtosis help in:
 - Identifying **non-normality**.
 - Deciding the **appropriate statistical test** (parametric vs non-parametric).

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